

Python Introduction

Motivation

Aug 2026	Aug 2025	Change	Programming Language		Ratings	Change
1	1			Python	18.53%	-7.61%
2	3	⬆		C	11.10%	+2.07%
3	2	⬇		C++	8.62%	-0.56%
4	4			Java	8.25%	-0.34%
5	5			C#	4.09%	-1.43%
6	6			JavaScript	2.63%	-0.52%
7	7			Visual Basic	2.18%	-0.15%
8	12	⬆		SQL	1.88%	+0.16%
9	14	⬆		R	1.56%	+0.19%
10	18	⬆		Rust	1.45%	+0.31%
11	10	⬇		Delphi/Object Pascal	1.37%	-0.46%

<https://www.tiobe.com/tiobe-index/>

Web development is also easy with Python

Start a simple webserver:

```
# python3 -m http.server
```

<https://pythonbasics.org/webserver/>

```
# Python 3 server example
from http.server import BaseHTTPRequestHandler, HTTPServer

hostName = "localhost"
serverPort = 8080

class MyServer(BaseHTTPRequestHandler):
    def do_GET(self):
        self.send_response(200)
        self.send_header("Content-type", "text/html")
        self.end_headers()
        self.wfile.write(bytes("<html><head><title>https://pythonbasics.org</title></head>", "utf-8"))
        self.wfile.write(bytes("<p>Request: %s</p>" % self.path, "utf-8"))
        self.wfile.write(bytes("<body>", "utf-8"))
        self.wfile.write(bytes("<p>This is an example web server.</p>", "utf-8"))
        self.wfile.write(bytes("</body></html>", "utf-8"))

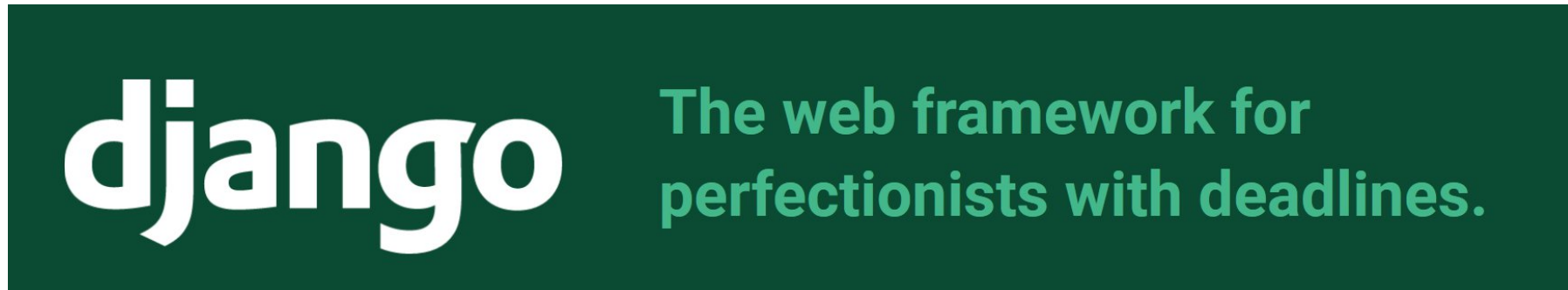
if __name__ == "__main__":
    webServer = HTTPServer((hostName, serverPort), MyServer)
    print("Server started http://%s:%s" % (hostName, serverPort))

    try:
        webServer.serve_forever()
    except KeyboardInterrupt:
        pass

    webServer.server_close()
    print("Server stopped.")
```

Lots of libraries and frameworks exist

For instance:



```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello_world():
    return "<p>Hello, World!</p>"
```

Python is widely used in industry

Instagram - <https://instagram-engineering.com/web-service-efficiency-at-instagram-with-python-4976d078e366>

Instagram currently features the world's largest deployment of the Django web framework, which is written entirely in Python.

Google - <https://stackoverflow.com/questions/2560310/heavy-usage-of-python-at-google/2561008#2561008>

Spotify - <https://engineering.atspotify.com/2013/03/how-we-use-python-at-spotify/>

It all got started, I believe, because the very earliest Googlers (Sergey, Larry, Craig, ...) made a good engineering decision: "Python where we can, C++ where we must."

Spotify's backend consists of many interdependent services, connected by [its] own messaging protocol over ZeroMQ. Around 80% of these services are written in Python.

Data science and ML is easy with Python

Simple example (155 line of code)

From: https://scikit-learn.org/stable/auto_examples/classification/plot_classifier_comparison.html

To run the example, make sure that you use Python 3:

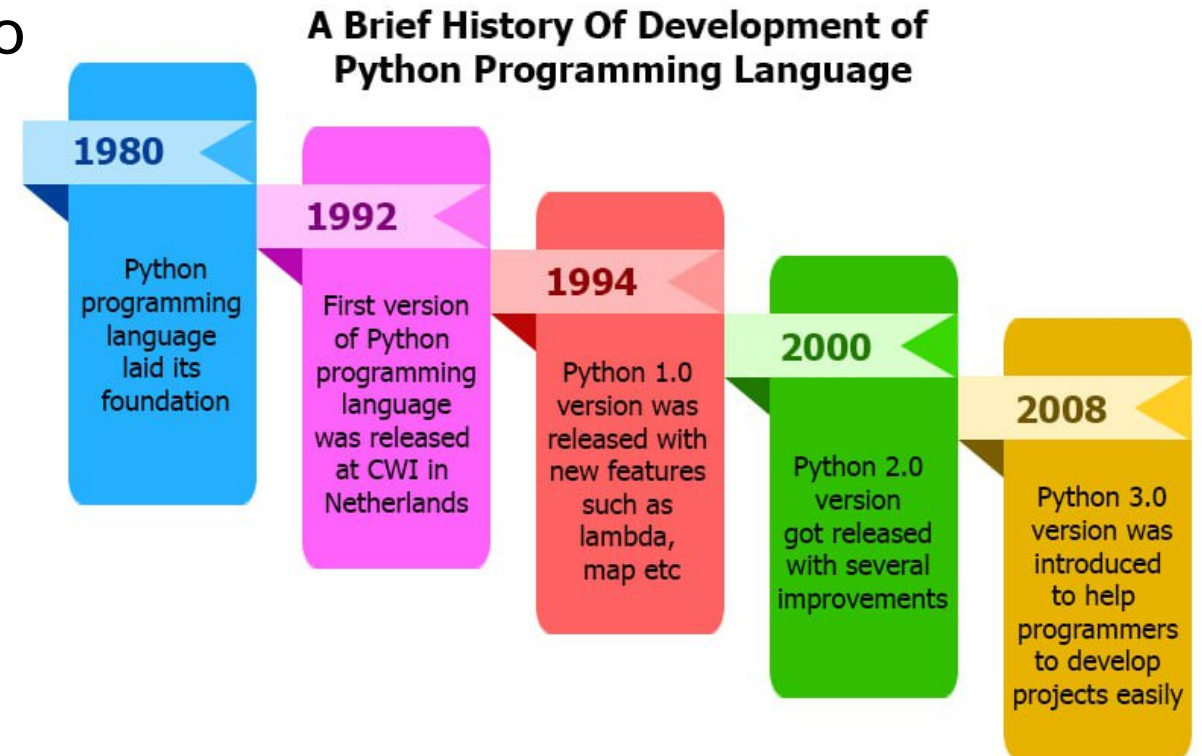
```
# pip3 install scikit-learn matplotlib
...
# python3 plot_classifier_comparison.py
```

Lots of libraries and frameworks exist

- SciPy provides algorithms for optimization, integration, interpolation, eigenvalue problems, algebraic equations, differential equations, statistics and many other classes of problems.
- Scikit-learn features various classification, regression and clustering algorithms including support-vector machines, random forests, gradient boosting, k-means and DBSCAN, and is designed to interoperate with the Python numerical and scientific libraries NumPy and SciPy.
- TensorFlow is an end-to-end open source platform for machine learning. It has a comprehensive, flexible ecosystem of tools, libraries and community resources that lets researchers push the state-of-the-art in ML and developers easily build and deploy ML powered applications. Developed by Google.
- Keras is an API designed for human beings, not machines. Keras follows best practices for reducing cognitive load: it offers consistent & simple APIs, it minimizes the number of user actions required for common use cases, and it provides clear & actionable error messages.

History

- Invented in the Netherlands by Guido Van Rossum
- Implementation was started in December 1989
- Named after Monty Python
- The creator of Python, Guido Van Rossum, was a fan of Monty Python's



Source: <https://www.facebook.com/Asterisminfosoft/photos/a.475015593349486/814876639363378/?type=3>

Language

“Python is a high-level, interpreted, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. Python is dynamically-typed and garbage-collected.” from Wikipedia

```
1  n = int(input('Type a number, and its factorial will be printed: '))
2
3  if n < 0:
4      raise ValueError('You must enter a non-negative integer')
5
6  factorial = 1
7  for i in range(2, n + 1):
8      factorial *= i
9
10 print(factorial)
```

Libraries

- Python's has a large standard library. It includes many standard formats and protocols such as MIME and HTTP for internet applications, modules for creating graphical user interfaces, connecting to relational databases, generating pseudorandom numbers, arithmetic with arbitrary-precision decimals, manipulating regular expressions, and unit testing.
- On top of the standard library, Python has the third-party Python Package Index with close to 883.000 packages (September 2026). Check: <https://pypi.org>

```
pip install <package-name>
```

How to install python

- Linux package, e.g. “`sudo apt install python3`” on Debian-based systems.
- Windows and Mac: download binaries from python.org
- After installation, you should be able to type: “`python --version`” and the output should be similar to:

```
C:\Users\alc>python --version  
Python 3.10.5
```

```
alc@surface:~$ python --version  
Python 3.8.10
```

- An alternative is Anaconda (or Miniconda) that makes package management and virtual environments much easier to set up

How to run

- `python <enter>` (or `python3 <enter>`)
- `ipython <enter>`
- `python <script_name>.py`
- jupyter notebooks
- IDE, e.g. pycharm, vscode, ...

...

Jupyter notebook

Online

- Google Colab: go to <https://colab.research.google.com>
- (and many others)

On your own computer:

```
pip install jupyter notebook
```

(If you get warnings about some files not being in your PATH, you may have to add the installation directory indicated to your PATH environment variable and restart your terminal/cmd)

```
cd <folder with your .ipynb file>  
jupyter notebook
```

Today's exercises

- Install Python and write the obligatory first “Hello World” program. Find some way to run it.
- Go through the notebook for today's lesson -- either upload it to a jupyter notebook site such as Google Colab, or run it on your own computer.
- More about jupyter notebooks: you can find a ton of tutorials online, for instance: <https://realpython.com/jupyter-notebook-introduction/>

Recent Linux releases (e.g. Ubuntu)

Python is used by system packages, hence the following won't work:

```
# pip3 install scikit-learn matplotlib
```

Instead, either install packages via system package manager (`apt install python3-matplotlib`) or use a virtualenv.

```
# sudo apt install python3-virtualenv
# python3 -m venv .venv (to create the virtual environment)
# source .venv/bin/activate (to enter the virtual environment)
# pip install [your packages...]
# python [run whatever]
# deactivate (to exit the virtual environment)
```